Proof

Let $\vec{a}$ and $\vec{b}$ be two vectors. Let $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$.

Complete the parallelogram $OACB$ with $\vec{a}$ and $\vec{b}$ as adjacent sides. The vectors $\overrightarrow{OB}$ and $\overrightarrow{AC}$ have same direction and equal magnitude; so $\overrightarrow{OB} = \overrightarrow{AC}$. Thus

$$\vec{a} + \vec{b} = \overrightarrow{OA} + \overrightarrow{AC} = \overrightarrow{OC}.$$

As, $\overrightarrow{OA} = \overrightarrow{BC}$,

$\vec{b} + \vec{a} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OC}$.

Thus $\vec{a} + \vec{b} = \vec{b} + \vec{a}$.

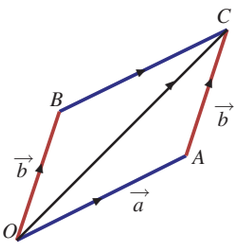


Fig. 8.21

Polygon law of addition

Let $\overrightarrow{OA}, \overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CD}$, and $\overrightarrow{DE}$ be any five vectors as shown in the Fig. 8.22.

We observe from the figure that each vector is drawn from the terminal point of its previous one. By the triangle law,

$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$; $\overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OC}$

$\overrightarrow{OC} + \overrightarrow{CD} = \overrightarrow{OD}$; $\overrightarrow{OD} + \overrightarrow{DE} = \overrightarrow{OE}$

Thus $\overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{OE}$.

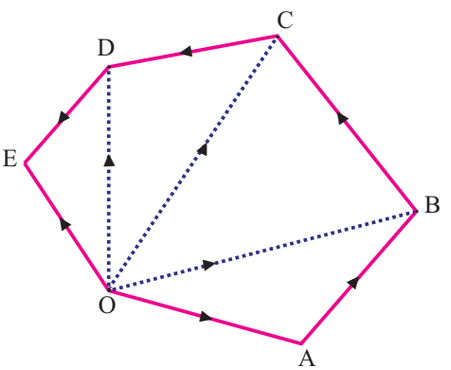


Fig. 8.22

Thus the line joining the initial point of first vector to the terminal point at the last vector is the sum of all the vectors. This is called the **polygon law of addition** of vectors.

Example 8.1

Represent graphically the displacement of

(i) 30 km 60° west of north

(ii) 60 km 50° south of east.

Solution

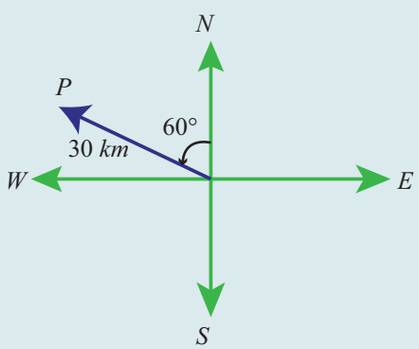


Fig. 8.23

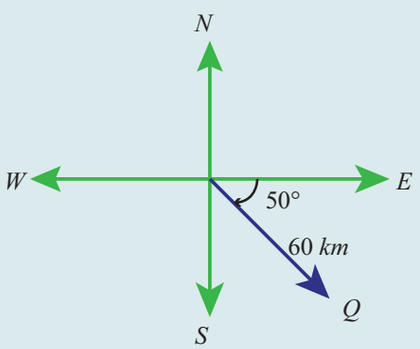


Fig. 8.24

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